# Glossary

## dating

Placing a sample or event in time.

## chronology

The temporal ordering of evidence and events.

## relative age

An older-or-younger relationship to other evidence.

## numerical age

An age expressed numerically, for example in years.

## calendar age

An age placed on a calendar timescale.

## temporal resolution

The time interval resolved by a record.

## uncertainty

The limited certainty of a measurement or inference.

## radiocarbon dating

Dating based on radiocarbon in a sample.

## calibration

Relating radiocarbon ages to calendar ages using a calibration curve.

## half-life

The time for half the atoms of a radioactive isotope to decay.

## stratigraphy

The study of strata and their relationships.

## luminescence dating

Dating time since mineral signals were reset by light or heat.

## palaeomagnetism

Records of past geomagnetic fields preserved in materials.

## dendrochronology

The study of change using calendar-dated tree rings.

## proxy

Evidence used to infer an environmental variable indirectly.

## paleoclimatology

The study of past climates.

## varve

A sedimentary layer verified to represent one year.

## vascular cambium

Meristem producing secondary xylem and phloem.

## xylem

Tissue involved in water transport and support.

## phloem

Tissue transporting products of photosynthesis.

## tracheid

A principal conducting and supporting cell of conifer wood.

## vessel

A water-conducting structure composed of vessel elements.

## ray

Radially oriented tissue in wood.

## earlywood

Wood formed in the earlier part of a growth increment.

## latewood

Wood formed in the later part of a growth increment.

## tree-ring boundary

The boundary between adjacent annual rings.

## growth ring

A ring-like increment associated with a growth cycle.

## sapwood

The outer, physiologically active portion of wood.

## heartwood

Inner wood transformed from sapwood, usually no longer conducting water.

## false ring

An intra-annual feature that resembles an annual boundary.

## locally absent ring

An annual ring absent from part of a stem.

## crossdating

Matching ring patterns to establish and verify ring dates.

## limiting factor

An environmental or physiological factor limiting growth.

## reference chronology

A dated chronology used as a reference for crossdating.

## floating chronology

An internally dated series not yet anchored to calendar years.

## ring width

The radial width of an annual ring.

## standardization

Processing growth trends and scales for the purpose of comparison.

## ring-width index

Ring width expressed relative to an estimated growth trend.

## dendroarchaeology

The study of past human activity using tree rings.

## felling year

The year in which a tree was cut down.

## dendroecology

The study of ecological processes using tree rings.

## growth release

An increase in growth, for example after reduced competition.

## disturbance

An event or process altering forest structure or growth.

## dendroclimatology

The study of climate variability using tree rings.

## correlation

A statistical association between variables.

## regression

Modelling a relationship between variables.

## validation

Checking model performance using independent information.

## stable isotope

An isotope that does not undergo radioactive decay.

## cell

A basic unit of biological structure and function.

## tissue

An organized group of cells with related functions.

## organ

A functional structure composed of several tissues.

## meristem

A region of dividing cells contributing to plant growth.

## primary growth

Growth increasing the length of stems or roots.

## secondary growth

Growth increasing thickness through lateral meristem activity.

## cell differentiation

Development of specialized cell structure and function.

## cell enlargement

Increase in the size of a developing cell.

## secondary cell wall

Wall deposited inside the primary wall, often after cell expansion.

## lignification

Deposition of lignin in cell walls.

## parenchyma cell

A cell involved in storage and metabolism, including living cells in wood.

## ring-porous wood

Wood with large vessels concentrated near the start of a growth ring.

## diffuse-porous wood

Wood with vessels more evenly distributed in size through a growth ring.

## transverse section

A section approximately perpendicular to the stem axis.

## radial section

A longitudinal section along a stem radius.

## tangential section

A longitudinal section approximately tangential to the growth rings.

## photosynthesis

Production of organic compounds using light energy and carbon dioxide.

## respiration

Metabolic processes releasing energy from organic compounds.

## transpiration

Loss of water vapour from plants, especially through stomata.

## stoma

An opening regulating gas exchange and water loss.

## carbon allocation

Partitioning acquired carbon among growth, storage and maintenance.

## phenology

The seasonal timing of events such as leaf-out and dormancy.

## dormancy

A state of temporarily suspended or reduced development.

## physiology

The study of biological functions and processes.

## ecology

The study of relationships among organisms and their environment.

## population

Individuals of the same species in a defined area.

## community

Populations of different species occurring together.

## ecosystem

Organisms and the abiotic environment interacting as a system.

## competition

Interactions involving access to limited resources.

## canopy gap

An opening in the forest canopy following events such as tree death.

## ecological succession

Change in community composition and structure through time.

## biotic and abiotic factors

Biological influences such as competition, and physical influences such as temperature and moisture.